

COMSATS UNIVERSITY ISLAMABAD

Attock Campus



Department Of Computer Science

Course	Mobile Application & Development
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City Explorer

Advanced React Native Map & Geofencing Application

Introduction:

City Explorer is a React Native mobile application developed using Expo and React Native Maps. The application provides real-time user location tracking, interactive map visualization, geofencing functionality, and nearby places detection.

The app allows users to:

- View their live location on a map
- Explore nearby points of interest
- Search locations worldwide
- Switch between different map types
- Detect nearby locations using geofencing
- View detailed information about selected places

The project demonstrates the implementation of advanced map components and modern mobile UI design using React Native and Expo.

Technologies Used

- React Native
- Expo
- React Native Maps
- Expo Location
- OpenStreetMap API
- Ionicons

Features

1. Live User Location Tracking

- Fetches current GPS location using Expo Location
- Displays real-time coordinates
- Shows animated user marker on the map

2. Interactive Map

- Displays map using React Native Maps

- Supports:
 - Standard Map
 - Satellite Map
 - Hybrid Map

3. Custom Markers

- Multiple custom markers are displayed for:
 - Hospital
 - Park
 - Mosque
 - University
 - Bus Terminal
 - Food Street

4. Geofencing System

- Detects nearby locations inside geofence radius
- Displays alerts when user enters nearby area
- Visual geofence circles are shown on map

5. Search Functionality

- Search any location worldwide
- Uses OpenStreetMap Nominatim API
- Displays searched place with marker

6. Distance Calculation

- Calculates distance between user and selected place
- Displays distance in kilometers

7. Animated User Interface

- Animated user pulse effect
- Sliding detail panel
- Interactive floating controls

8. Polyline Navigation

- Draws route line between user and selected location

Components Used

Component	Purpose
MapView	Displays map
Marker	Shows locations on map
Circle	Displays geofence area
Polyline	Shows route path
Callout	Displays place details
Animated	Creates UI animations

Working of Application

Step 1: Location Permission

- Application requests user location permission
- If permission is denied, error message is displayed

Step 2: Fetch Current Location

- User's current GPS coordinates are fetched
- Map centers automatically on user location

Step 3: Display Map & Markers

- MapView displays the map
- User marker and nearby place markers are shown

Step 4: Geofencing Detection

- Application checks distance between user and nearby places
- If user enters geofence radius:
 - Alert notification appears
 - Nearby place is highlighted

Step 5: Search Places

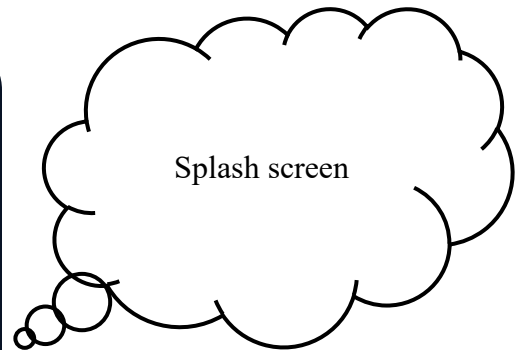
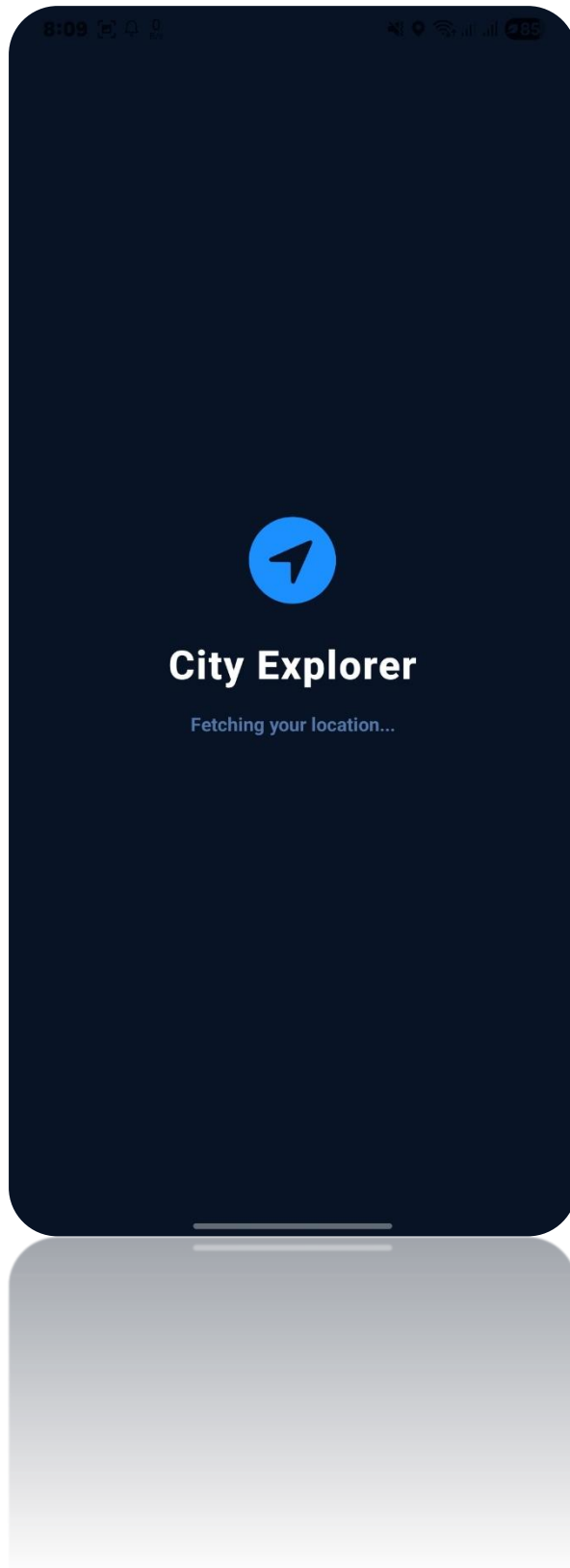
- User can search any location
- Search results are fetched using OpenStreetMap API
- Selected result is marked on map

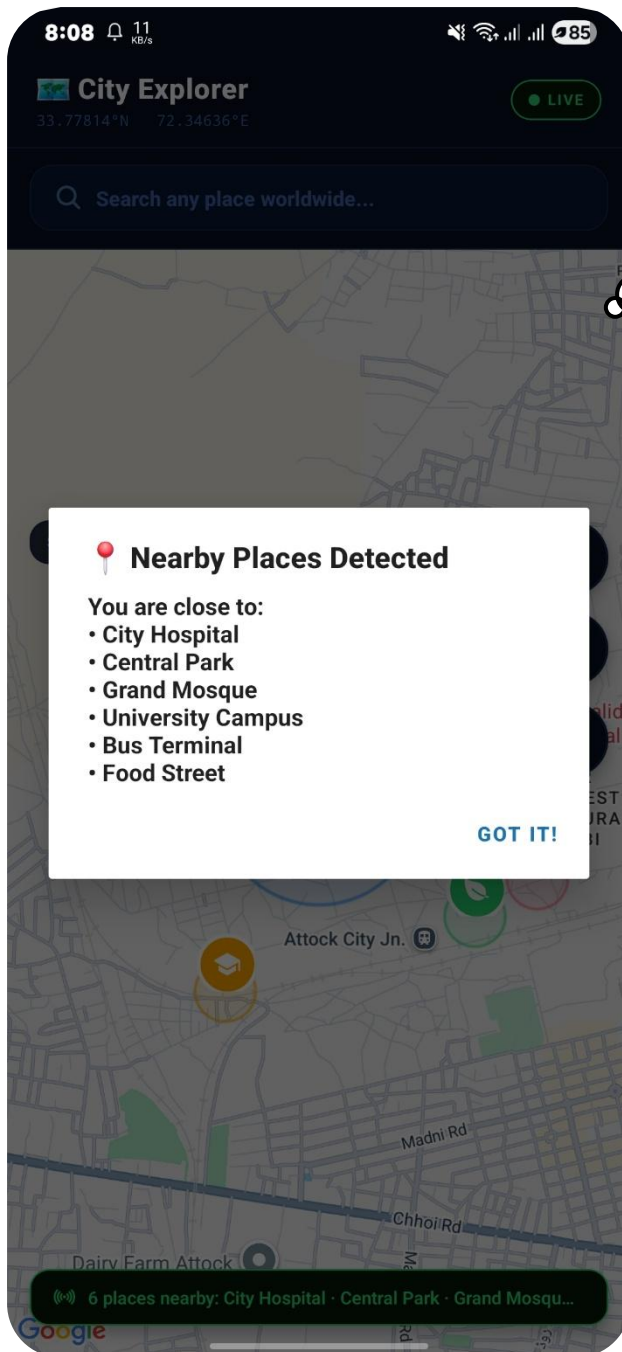
Step 6: Display Place Details

- When marker is selected:
 - Detail panel opens

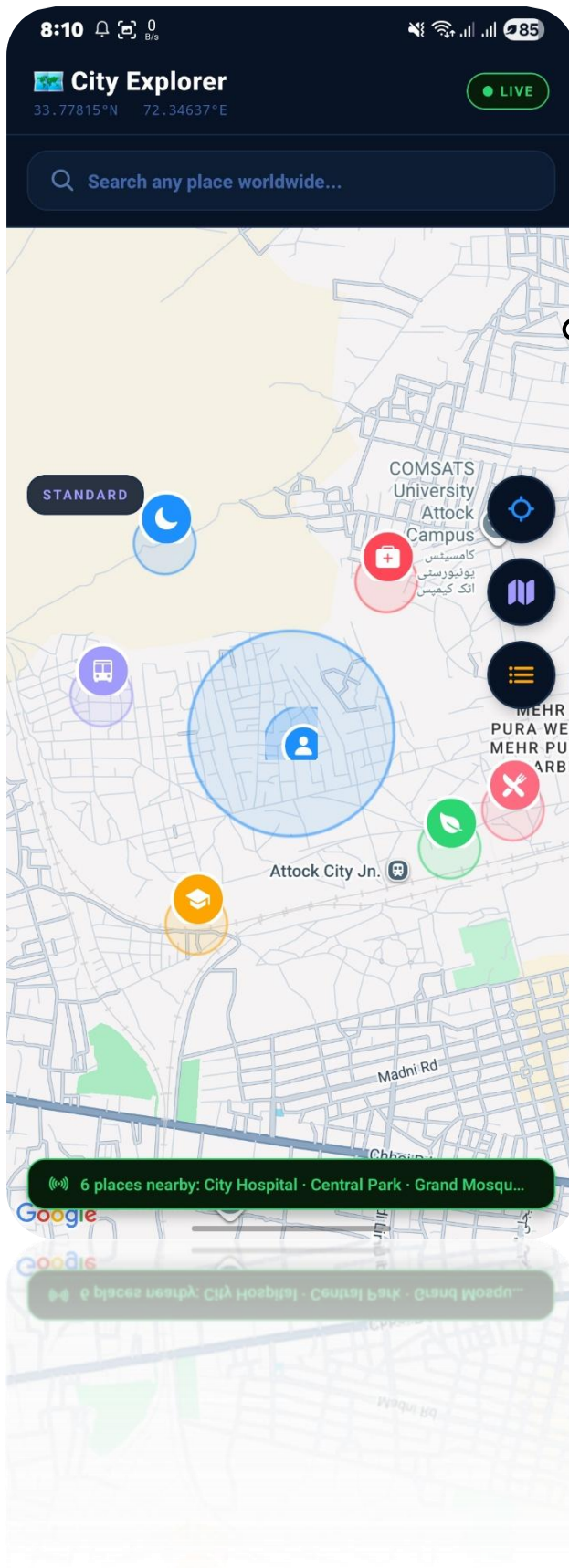
- Distance information is displayed
- Route line is drawn

Screenshots

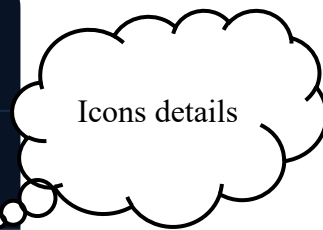
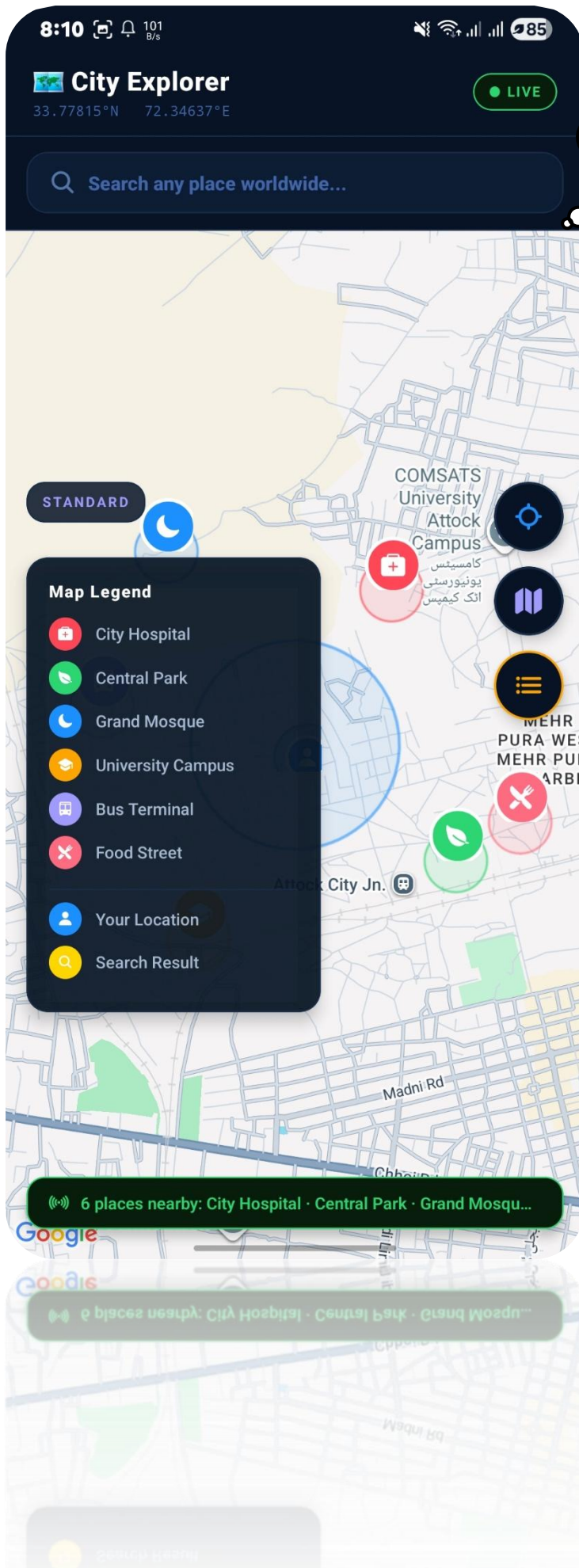


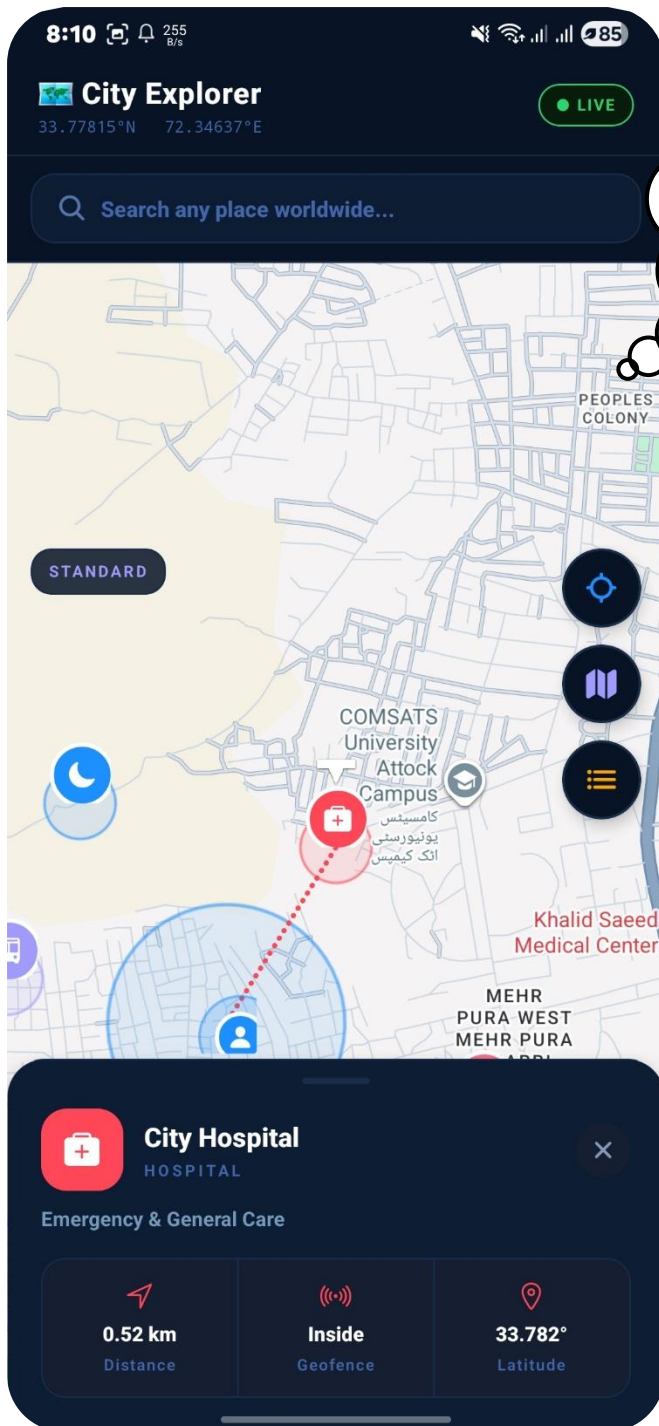


After Getting permissions of location from user, this shows

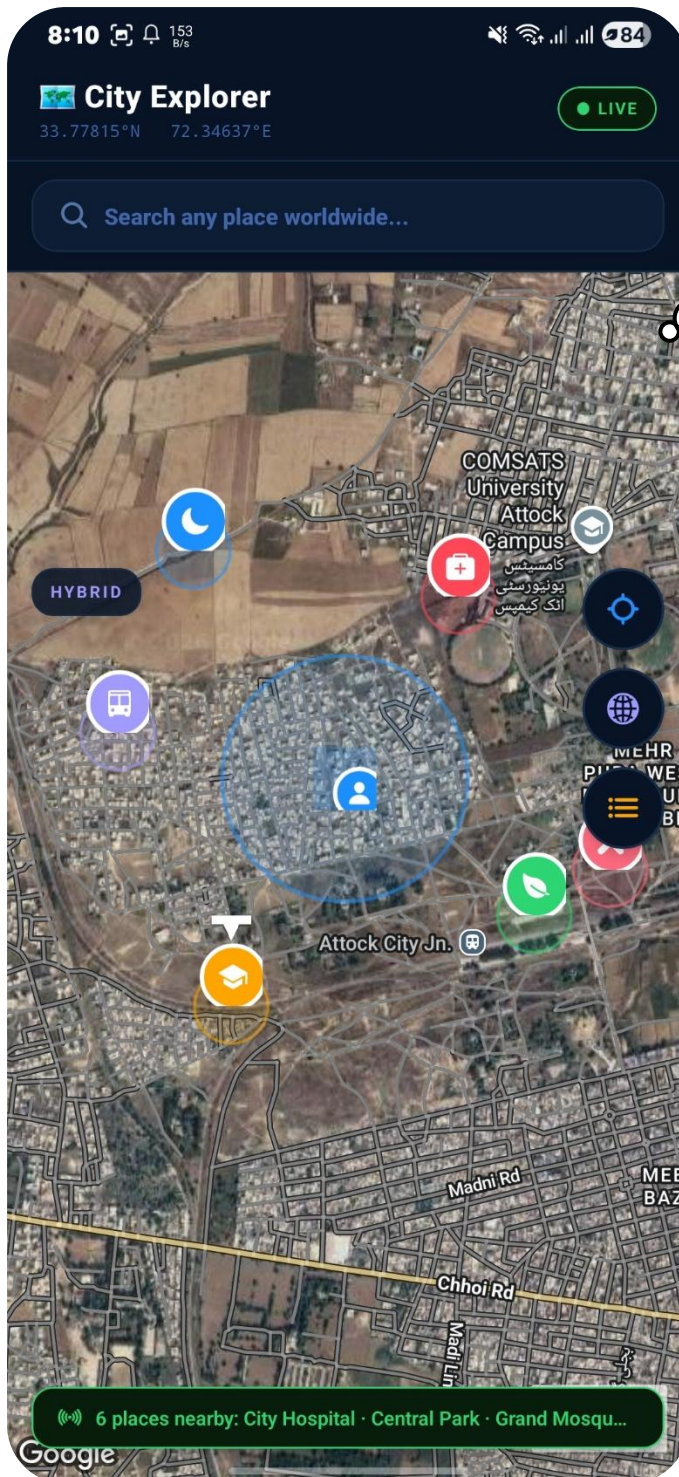


Showing my current location and I use different icons to represent different places

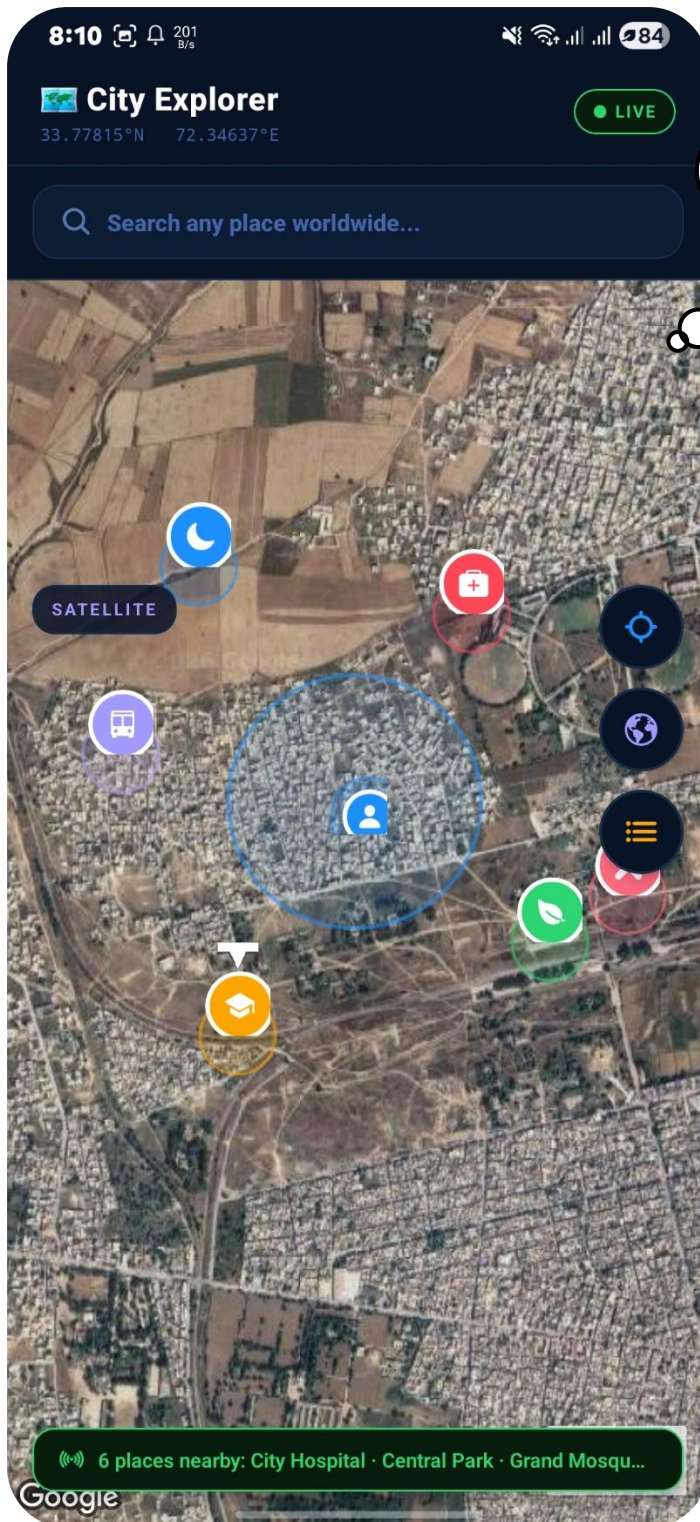




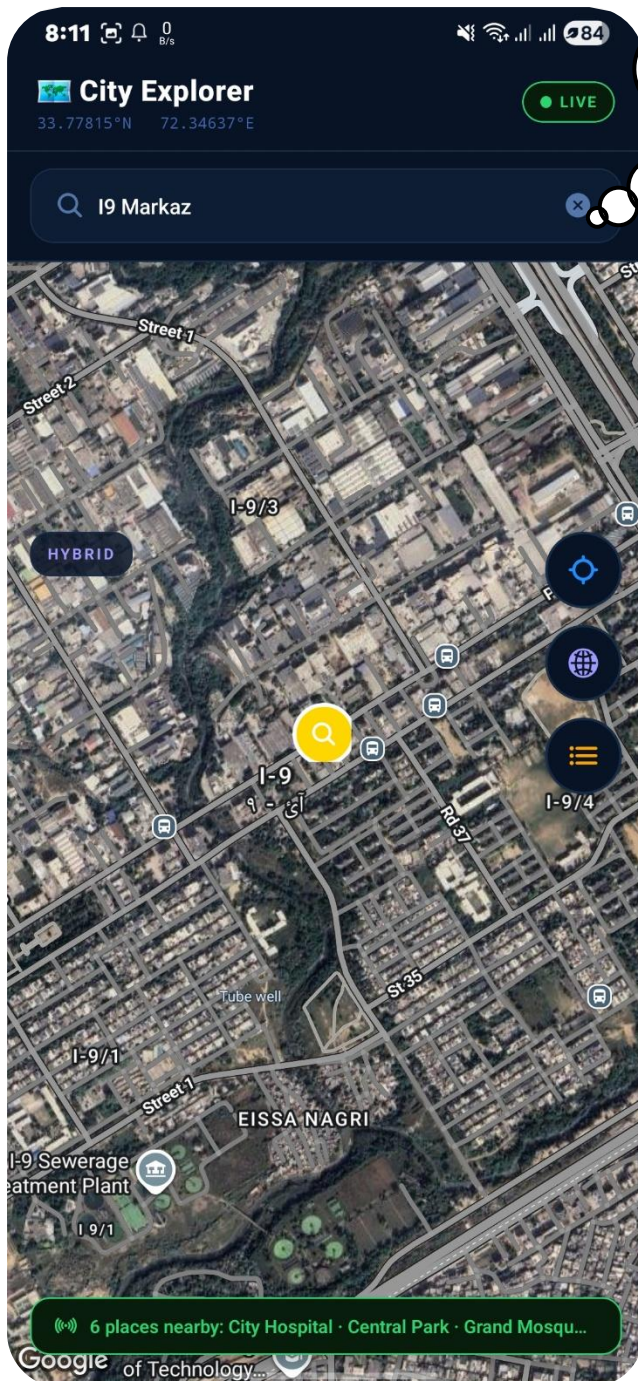
Proper showing Info
when click on any
icon



Previous was Standard
mode, now this is
Hybrid Mode
(Standard + Satellite)

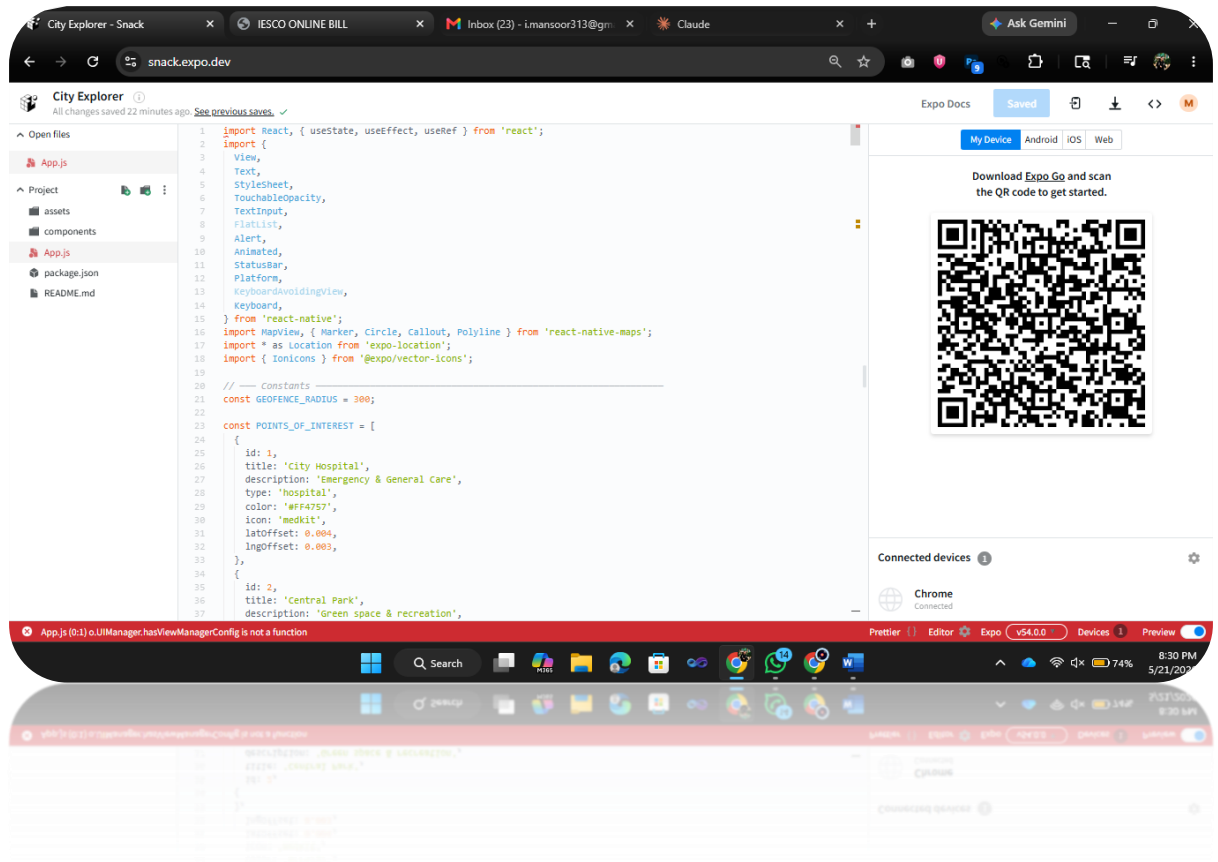


This is only satellite mode



We can search any location globally and it properly works

Code Screenshot:



Note About Error in Snack Web Preview

Explanation

While running the project in Expo Snack, a red-line error appears in the Web Preview:

```
UIManager.hasViewManagerConfig is not a function
```

This issue occurs because `react-native-maps` has limited support in Expo Snack Web mode.

However, the application works correctly in:

- Expo Go App
- Android Device
- iOS Device

The project was successfully tested on Expo Go mobile application, where all functionalities such as:

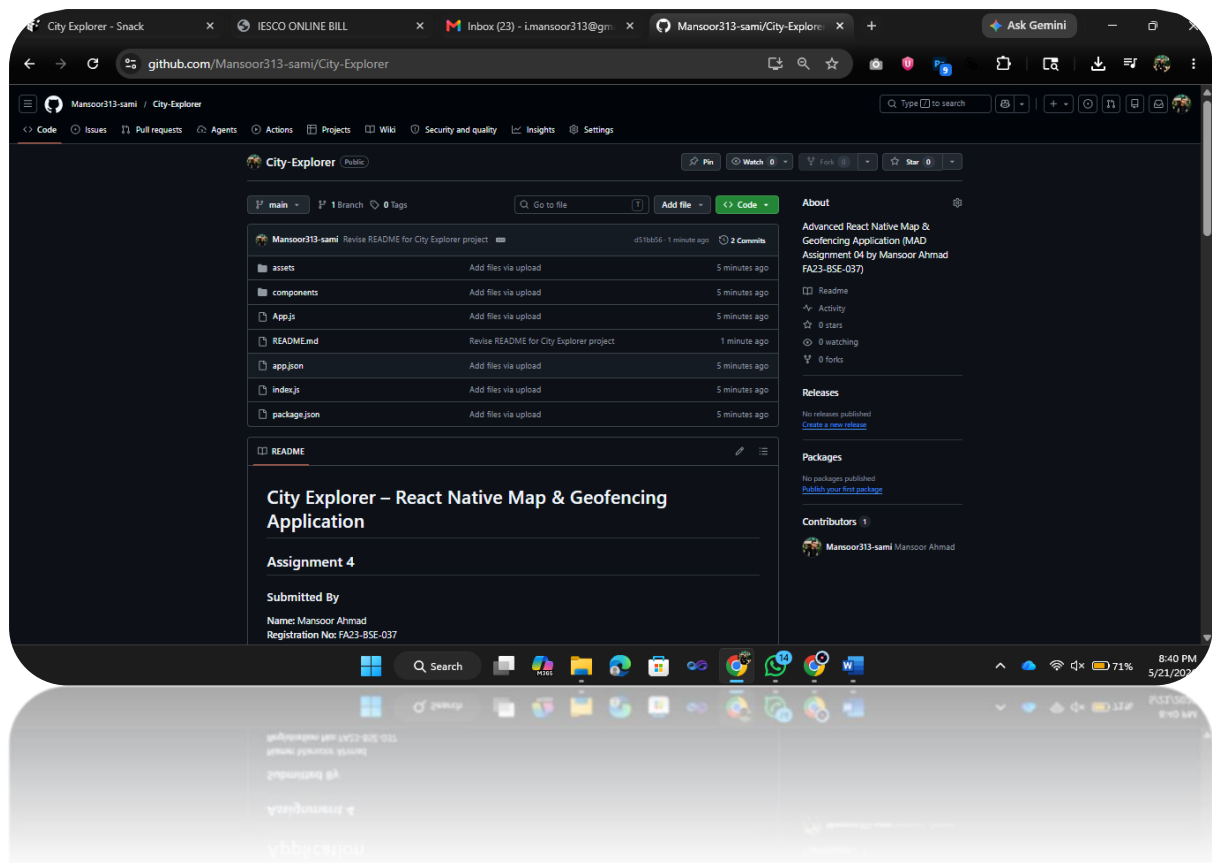
- Map rendering

- User location
- Markers
- Geofencing
- Search functionality
- Route polyline

worked properly without any issues.

My Repository Link:

<https://github.com/Mansoor313-sami/City-Explorer.git>



Thanks !